UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Data representation and number systems

Convert -20.125 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

Answer: -20.12510 = 0xC1A10000 (sign=1, exponent=10000011, fraction=01000010000000000000000)

- *1 mark for the sign bit.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the correct 32-bit grouping.*
- *1 mark for the final hexadecimal value.*

QUESTION A2 (5 marks)

Topic: Computer generations and history

Question 1 is based on the following statement.
Mr. X is known as Father of Computer. He developed Analytical Machine in 1837 which has four main components, namely; memory, the mill, punch card reader and the punch and printed section. Analytical Machine is the successor of Y machine. Y machine is used to compute tables of numbers for naval navigation.

a)
Name the inventor of Mr. X.

b)
Identify the name of Y machine.

c)
What is the function of Analytical Machine?

d)
Map the components Analytical Machine to Von Neumann architecture

Source: CSC429 (4).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Parallel and multiprocessor architecture

a)
Give ONE (1) characteristic of Symmetric Multiprocessor (SMP).

b)
Discuss the meaning of "symmetric" in Symmetric Multiprocessor.

c)
Give ONE (1) advantage of Symmetric Multiprocessor organization.

Source: CSC429 (10).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2

Label the addressing technique illustrated in Figure 2.

a)

Explain how the addressing technique in 9(a) is used.

b)

List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Computer architecture and CPU organisation

a) Describe the function of each computer bus type.

b) Discuss TWO (2) benefits of synchronous bus timing.

Source: CSC429 (7).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Data representation and number systems

Answer both parts for the same expression. Assume an 8-bit byte.

Expression: 24 - (-83)

a) Calculate 24 - (-83) using sign and magnitude.

Answer: left 24 -> 0 0011000; right -83 -> 1 1010011; 24 - (-83) = 107 -> sign-magnitude 0 1101011

- *1 mark for operand representations.*
- *1 mark for sign handling.*
- *1 mark for magnitude arithmetic.*
- *1 mark for the result sign bit.*
- *1 mark for the final byte.*

b) Calculate 24 - (-83) using 1's and 2's complement.

Answer: left 24 -> 1's 00011000, 2's 00011000; right -83 -> 1's 10101100, 2's 10101101; 24 - (-83) = 107 -> 1's 01101011, 2's 01101011

- *1 mark for the positive operand in binary.*
- *1 mark for the 1's complement of the negative operand.*
- *1 mark for adding 1 to form the 2's complement.*
- *1 mark for the correct 8-bit result.*
- *1 mark for converting the result back to decimal.*

QUESTION A7 (10 marks)

Topic: Computer architecture and CPU organisation

Processor contains a number of registers used for temporary storage of data. Describe FIVE (5) types of registers.

Source: CSC429 (2).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Input/output and interrupts

a)

Define interrupt.

b)

Differentiate between interrupt and direct manipulation access.

c)

Given the following scenario.

Dilan is surfing woolworth.com to buy groceries online. While he is comparing prices of products in woolworth.com and one other website, two messages popped up onto his screen. Figure 2 (a) and (b) illustrate the conveyed messages.

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Are you sure you want to
There is a hardware event failure
in this computer. Please reboot
cancel this request?
this computer now.
(a)
(b)
Figure 2: The pop-up messages

i)
Identify the type of interrupt in Figure 2(a) and Figure 2(b) with proper justification.

ii)
While interruptroutinefor Interrupt Ais executing, it is interrupted by interrupt Bwhich has a higher priority than interrupt A. What will happen to interrupt routine A?

Source: CSC429 (13).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Input/output and interrupts

a) Why input/output problem occurs?
b) Describe THREE (3) categories for input/output module and provide an appropriate example each.

Source: CSC429 (6).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Memory hierarchy and cache

a) Rank the following memory types based on price ranging from the cheapest to the most expensive.
Cache, tape back-up, register, hard drive
b) Discuss ONE (1) benefit of cache memory.
c) Differentiate between write through and write back mechanisms in order to store data in cache.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 2 and Table 1.

y = A -
2C
* D
B + A2

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three-address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

Three-address	One-address	Two-address
LOADp		
MOV (p,q)		
ADD (p.q.r)		
STORp		
ADD (p,q)		
SUB (p,q,r)		
ADDp		
SUB (p,q)		
MUL (p.q.r)		
SUBp		
MUL (p,q)		
DIV(p,q,r)		
MULp		
DIV(p,q)		
DIVp		

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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